



UTM
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SBEU3213 FIELD ASTRONOMY

INDIVIDUAL REFLECTION

SECTION EB1

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I declare that this reflection is my own work and is based on my personal experiential learning

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S - SITUATION

The experiential learning activity was an astronomy service learning program called The Cosmic History as part of the Field Astronomy course. The main purpose of this program was to share basic knowledge about the cosmos, the universe and any topic that related to astronomy with upper form students from secondary school, while allowing us as a university students to apply theoretical knowledge learned in the class to a real world educational situation. Through this program, students were given the opportunity to learn from the astronomy talk and also enjoy the creative activity that could keep them entertained while learning something new about astronomy around us.

The program, Cosmic History took place at Sekolah Menengah Kebangsaan Skudai on 19 December 2025 and involved various participants, including around 40 students of a combined age of 15 and 16 years old. The session was organized in a structured manner, beginning with an introduction to astronomy followed by four entertainment activities including creating their own creative planet, arranging the solar system using balloons, throwing ping pong balls with answering some questions and lastly guessing the astronomy terms. These activities provided a meaningful learning experience by combining academic knowledge with interaction among students or facilitators, allowing students to exchange some ideas and cultivate interest in astronomy that can be useful as a STEM knowledge.

T -TASK

My role in this astronomy service learning program was to support the planning and delivery of the astronomy talk for upper secondary students. The assigned topic that I have to make for an astronomy talk is about how planets were created. I was required to prepare content related to the cosmos and the universe, assist in developing presentation materials, and help explain key concepts during the session. In addition, I was responsible for creating an activity that can attract students' attention on learning about astronomy. Then I automatically become a lead to that activity like give a do's and don't's as well as how much people can participate in the activity. Lastly, I was assigned to create a question with the answer that has three difficulties including easy, medium and hard. The importance of this role so that the students can brainstorming their ideas to answer the question.

A - ACTION

To carry out my role in the astronomy service learning program, I actively participated in the preparation and also gave some ideas about activities, specifically activity what we called "Kad Kosmik". A brief explanation about this activity is that students must guess exactly what terms on their head with the times given, a correct guess will count as correct and get a mark. With the same activity, I contributed to make a question slide that follows that astronomy talk earlier. I also prepare a presentation that slides by simplifying complex concepts related using diagrams, key points, and clear explanations. I chose this approach to ensure that the content could be easily understood by students, especially since it is their first time learning about astronomy.

During the actual session, I explain the activity that it is my own idea to begin with and also do a demonstration to ensure that students get the purpose of this activity. After the talk from one of our members, I made a decision to ask the students a simple question so we know who is actually focused during the talk. This strategy helped create a more interactive learning environment and ensured that the objectives of the program were achieved.

R - RESULT

The astronomy service learning program was successfully conducted and the objectives of the activity were completely achieved. The students showed a great interest and participated in all the activities throughout the session and discussions, indicating that they understood all the contents presented. However, there is one challenge throughout the program, it was time management. The time given for holding a program was so short, therefore we plan to make astronomy talk at a faster pace. As a result, some students did not understand the presentation well and not to mention there were certain topics that required more explanation than initially planned. Despite all of this, this program had a positive impact for both the participants and our team, as it enhanced students' understanding of astronomy while also improving our confidence in delivering scientific knowledge.

R - REFLECTION

Through this experience, I learned that effective teaching is more than subject knowledge but it also involves clear communication, adaptability and patience. Explaining complex astronomy concepts to students who are just a secondary school student has made me more aware of my own understanding of the subject. So, I just need to simplify and organise information in a logical manner. This whole process helped strengthen my understanding of astronomical topics like the solar system, planets, stars, universe and cosmic phenomena. Not only that, it also improves my ability to communicate complex ideas into simple terms.

This experience is closely related to the course content, as it allowed me to apply theoretical knowledge learned in the Field Astronomy course to a real world educational setting. When applying the principles of this course to teach others about these topics, I found that using theoretical concepts learned from this class gave those ideas significantly more value. This concept of applied learning is discussed in the course. In future, I would do the same program but with the adjustments to more challenging activities to further enhance their knowledge and improve learning experience. Overall, I learned that we can use service learning programs to effectively turn our academic knowledge into an applied and meaningful form of education.